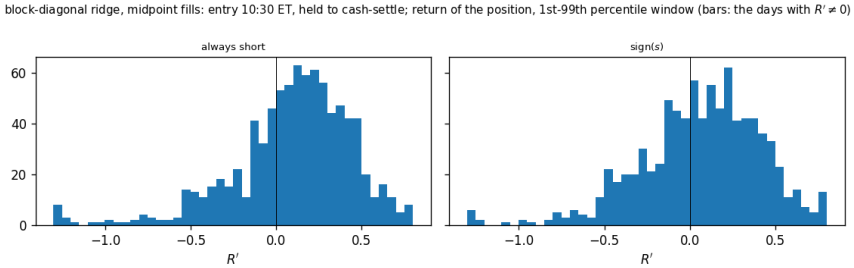
0DTE nearest-OTM straddle: per-rule return summary, entry 10:30 ET, held to cash-settle, models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.096	0.366	-2.00	0.88	-1.44	4.42	7.74	4.18	0	0.0	3.51
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.055	0.375	-1.81	2.00	-0.39	2.87	4.35	2.35	199	23.0	1.70
block-diagonal ridge	865	0.066	0.373	-2.00	1.91	-0.65	3.15	5.19	2.80	210	24.3	2.15
block-diagonal ridge, without the FOMC columns	865	0.066	0.373	-2.00	1.91	-0.64	3.15	5.23	2.82	207	23.9	2.17
LightGBM†	865	0.066	0.373	-1.81	2.00	-0.38	2.96	5.20	2.81	197	22.8	2.15
XGBoost†	865	0.069	0.372	-1.81	2.00	-0.39	3.00	5.46	2.95	193	22.3	2.29
lasso (causally tuned)†	865	0.063	0.373	-2.00	1.91	-0.72	3.16	4.95	2.67	201	23.2	2.02
lasso ($\alpha = 10^{-4}$)	865	0.064	0.373	-2.00	1.91	-0.64	3.12	5.04	2.72	201	23.2	2.06
elastic net (causally tuned)†	865	0.074	0.372	-2.00	1.91	-0.64	3.24	5.82	3.14	198	22.9	2.48



Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's thirteen overlapping remaining-session books. That reserved name is the book's daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar's share, and h_t is the chain's hours to the close.